

Vishnu Alachi

Software Engineer | Full-Stack & AI-Driven Systems

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SUMMARY

Software Engineering student (Cal Poly, B.S., expected Dec. 2026) with production experience architecting scalable, fault-tolerant automation platforms, distributed backend systems, and full-stack web applications. Oracle Cloud Infrastructure Certified Generative AI Professional.

EDUCATION

California Polytechnic State University

San Luis Obispo, CA

B.S. in Software Engineering | GPA: 3.4

Expected December 2026

- Relevant Coursework: Data Structures, Design & Analysis of Algorithms, Artificial Intelligence, Deep Learning, Computer Vision, Object-Oriented Design

SKILLS

Languages & Frameworks: Java, Python, JavaScript, TypeScript, C, C++, PL/SQL, React, Spring Boot, Node.js

Databases & DevOps: PostgreSQL, MongoDB, Oracle SQL, SQLite, Git, Jenkins, Docker, Agile/Scrum, CI/CD

AI & Systems: Codex/AI Agent IDEs, Custom Agent Skill Design, MCP (Model Context Protocol), AutoGen, RAG, Prompt Engineering, Computer Vision (OpenCV, YOLOv8), ROS2

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (Oracle, Sept 2025); LinkedIn Learning: AI Agents with AutoGen, AI Reasoning Models, Cybersecurity Fundamentals (July 2025)

EXPERIENCE

Oracle

Pleasanton, CA

Software Developer Intern

June 2026 – September 2026

- Architected a scalable merge automation platform, replacing a 20+ step manual workflow (APEX, BugDB, ADE, OraReview, PMC, Jenkins, Slack) with a fault-tolerant pipeline combining a deterministic state machine, SQLite-backed crash recovery, and a guarded LLM hybrid controller
- Benchmarked 5 LLM configurations across 20 test scenarios, achieving a 100% deterministic test pass rate and deploying the top-performing model at an 86.67% pass rate with zero safety violations

Oracle

Pleasanton, CA

Software Developer Intern

June 2025 – September 2025

- Built an internal developer platform (Codex plugin + Python MCP server) enabling natural-language automation of Jenkins/BugDB operations with signed async job tracking, plus a race-condition-safe round-robin load-balancing queue across 4 build pipelines
- Developed an AI agent harness (Python, AutoGen, React) with RAG-powered diagnostics, boosting response accuracy from 50% to 95% via prompt engineering

Hack4Impact

San Luis Obispo, CA

Software Developer

September 2024 – September 2025

- Rebuilt full nonprofit website (Spokes for Nonprofits) on a 9-person Agile team, automating a manual Google Form/Sheet job-posting workflow into a direct-to-database pipeline
- Owned end-to-end development of footer, skeleton-loading, and carousel filtering features in React, integrating MongoDB and role-based access control for secure authentication

Exalture Software Labs

Kochi, India

Software Development Intern

March 2023 – May 2023

- Optimized KNN model hyperparameters, improving evaluation accuracy for Python-based satellite anti-collision software

TECHNICAL PROJECTS

Poly Planner Pro (Senior Capstone Project) | Team of 5

September 2025 – Present

- Engineered a PDF-parsing pipeline extracting and structuring Cal Poly transcript data, cross-validating parsed courses against a live course-catalog API; deployed and actively used by Cal Poly's Academic Advising department for student course planning
- Implemented React-based advisor flowchart PDF export integrated with Spring Boot, Spring Data JPA, and PostgreSQL REST APIs

Dual-Arm Robotic Manipulation (Faculty Research Project) | Team of 2

January 2026 – Present

- Engineered a dual-mode control stack for a 6-DOF robotic arm (Waveshare RoArm-M3) in Python and ROS2, combining a low-level serial protocol with MoveIt2-based motion planning for Cartesian pick-and-place automation
- Implemented a geometric inverse-kinematics solver converting Cartesian targets to joint angles, with workspace-bounds and per-joint limit validation
- Built a real-time computer vision pipeline (YOLOv8, OpenCV) for object detection and pixel-to-world coordinate mapping to drive autonomous pick-and-place

Findr | Team of 4

April 2026 – June 2026

- Co-built a full-stack event-discovery platform (React, Node.js/Express, MongoDB) with JWT and Google OAuth authentication, RESTful APIs, and interactive map-based search; deployed to production on Azure with CI/CD via GitHub Actions
- Implemented Jest/Supertest integration tests and Cypress end-to-end acceptance tests across the application